SUMANTH M B

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SUMMARY

Computer Science Engineering student focused on Java backend development and distributed systems. Experience building REST APIs and Spring Boot microservices, event-driven pipelines with Apache Kafka, and data layers using MySQL, Redis, and Neo4j. Foundation in data structures and algorithms. Seeking backend engineering internship and full-time opportunities.

EDUCATION

Presidency University, Bengaluru | 2023 - 2027
B.Tech - Computer Science & Engineering (Data Science) | CGPA: 7.06
Class XII: 93% | Class X: 84%

TECHNICAL SKILLS

- Languages: Java 17 & 21, Python 3, SQL, JavaScript, HTML, CSS
- Backend & Frameworks: Spring Boot 3 & 4, Spring Data JPA, Microservices, REST API Design
- Databases & Messaging: MySQL, PostgreSQL, Redis, Neo4j, Apache Kafka
- Tools & DevOps: Docker, Docker Compose, Git & GitHub, Maven, Postman, Linux CLI
- Core CS: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, OOP

PROJECTS

Distributed Log Analytics & Microservices Monitoring Platform *GitHub*
Java 17, Spring Boot, Apache Kafka, Redis, Neo4j, Docker Compose, REST, HTML/CSS/JS

- Built 6 independent Spring Boot microservices (5 domain services and a dashboard service) with inter-service REST calls; each publishes structured log events (level, latency, timestamp) to an Apache Kafka topic, consumed in real time by a consumer-group listener
- Implemented Redis live counters through an atomic Lua script that de-duplicates by Kafka partition offset, giving idempotent per-service and per-minute INFO/ERROR counts; added health polling across 5 services
- Modeled the 5-service call graph in Neo4j and implemented weighted PageRank in Java (damping 0.85, 50 iterations) to rank bottleneck services; containerized the stack as a 10-container Docker Compose deployment (Kafka, ZooKeeper, Redis, 6 services, traffic generator) behind a live REST-driven dashboard
- Built a Docker-based traffic generator and chaos scripts (per-service failure-rate settings plus a payment-service outage toggle) to drive continuous load across all 5 services and reproduce failure scenarios

DPI Analyzer - Real-Time Deep Packet Inspection System *GitHub*
Java 17, Spring Boot 3.3, pcap4j, MySQL, Spring Data JPA, SSE, HTML/CSS/JS

- Engineered a real-time packet capture engine using pcap4j that inspects live network traffic across 9+ protocols (HTTP, HTTPS, DNS, SSH, MySQL), capturing IP, port, and transport-layer metadata per packet
- Designed an SSE-based streaming pipeline to push live traffic events to a browser dashboard with sub-second latency, eliminating client-side polling
- Built a bounded in-memory ring buffer using a Linked Blocking Deque to manage high-throughput packet streams without memory overflow under sustained load
- Integrated Spring Data JPA with MySQL for persistent traffic logging and historical packet query support across sessions
- Developed a responsive, dark-themed monitoring dashboard with real-time chart updates and protocol-level traffic breakdown

ACHIEVEMENTS

- Top 3 finish at TechBiz College Hackathon - built and demoed a functional prototype under a 24-hour time constraint
- Earned HackerRank certifications in Java and Python; completed multiple badge tracks in Problem Solving and SQL
- Coursera "Python for Everybody" Certification - foundational data programming and API integration